(b) $f(x) = e^{-x}(x-1)^{1/3}$

$f(-x) = e^{-(-x)}(-x-1)^{1/3} = e^{x}(-x-1)^{1/3} \neq f(x) \neq -f(x)$

$f(x)$ is neither even nor odd

(a) $f(x) = \sin(x)(\sin x - 1)$

$f(-x) = \sin(-x)\{\sin(-x) - 1\}$

$= -\sin(x)(-\sin x - 1) \neq -f(x)$ [not odd]

$= \sin x(\sin x + 1) \neq f(x)$ [not even]

## * Definition of Fourier Series:

Let, $f(x)$ be defined in the interval $(-L, L)$ and determined outside of this interval by $f(x+2L) = f(x)$ i.e., assume that $f(x)$ has period $2L$. The fourier series or Fourier expansion corresponding to $f(x)$ is defined to be:

$$\frac{a_0}{2} + \sum_{n=1}^{\infty}\left(a_n \cos\frac{n\pi x}{L} + b_n \sin\frac{n\pi x}{L}\right) \quad \text{---(1)}$$

Where the Fourier coefficients $a_n$ and $b_n$ are;

$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\frac{n\pi x}{L}\,dx \rightarrow n = 0, 1, 2, \ldots$$

$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\frac{n\pi x}{L}\,dx \rightarrow n = 1, 2, 3, 4, \ldots$$

$$a_0 = \frac{1}{L}\int_{-L}^{L} f(x)\,dx.$$

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# Integration Formula

1) $\int x^n dx = \frac{x^{n+1}}{n+1}$

2) $\int \sin(mx) dx = \frac{-\cos(mx)}{m} + c$

3) $\int \cos(mx) dx = \frac{\sin(mx)}{m} + c$

4) $\int (uv)\, dx = u\int v\, dx - \int \left(\frac{d}{dx}(u)\int v\, dx\right) dx$

# $\int x \sin 2x\, dx = x\int \sin(2x) dx - \int \left(\frac{d}{dx}(x)\int \sin(2x) dx\right) dx$

$$= -\frac{x\cos(2x)}{2} - \int 1 \cdot \left(\frac{-\cos 2x}{2}\right) dx$$

$$= \frac{-x\cos(2x)}{2} + \frac{\sin(2x)}{2} + c$$

Apply LIATE for choosing u and v.
L → Logarithmic, I → Inverse, A → Arithmatic,
T → Trigonometric, E → Exponential.

## Trigonometric Identities

1) $2\sin A \sin B = \cos(A-B) - \cos(A+B)$
2) $2\cos A \cos B = \cos(A+B) + \cos(A-B)$
3) $2\sin A \cos B = \sin(A+B) + \sin(A-B)$.
4) $2\sin^2 A = 1 - \cos 2A$
5) $2\cos^2 A = 1 + \cos 2A$

* $\sin(n\pi) = 0$
* $\cos(n\pi) = (-1)^n$
* $\sin(-n\pi) = 0$
* $\cos(-n\pi) = (-1)^n$

[for any integer n
$n \in N$ or $n \in Z$]

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## * Half Fourier Series:

* If $f(x)$ is even, $b_n = 0$.
* So, no sine term in fourier series, only cosine term → 'cosine series'.
* If $f(x)$ is odd, $a_n = 0$.
* So, no cosine term, only sine term → 'sine series'.

## * Even-Odd Function:

$f(-x) = f(x)$ : Even → Symmetric about y-axis

$f(-x) = -f(x)$ : Odd → Symmetric about origin (symmetric about bot y and x)

## * Calculating Fourier Series: $(-L, L)$:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[ a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right]$$

$$a_n = \frac{1}{L} \int_{-L}^{L} f(x) \cos\left(\frac{n\pi x}{L}\right) dx$$

$$b_n = \frac{1}{L} \int_{-L}^{L} f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

$$a_0 = \frac{1}{L} \int_{-L}^{L} f(x) dx .$$

* For $(-\pi, \pi)$ interval $L = \pi$:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[ a_n \cos(nx) + b_n \sin(nx) \right]$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$$

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## * Half Series $(0, \pi)$:

### Even Function ($b_n = 0$) — Cosine Series

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos(nx)]dx$$

$$a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos(nx)dx$$

$$a_0 = \frac{2}{\pi}\int_0^{\pi} f(x)dx$$

### Odd Function ($a_n = 0$) — Sine Series

$$f(x) = \sum_{n=1}^{\infty} [b_n \sin(nx)]$$

$$b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin(nx)dx$$

Q.1: $f(x) = \begin{cases} x & \text{for } 0 < x < \pi \\ -x & \text{for } -\pi < x < 0 \end{cases}$.

Find the Fourier series expansion of $f(x)$.

Sol$^n$: $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos(nx) + b_n \sin(nx)]$

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)dx$$

$$= \frac{1}{\pi}\int_{-\pi}^{0} (-x)dx + \frac{1}{\pi}\int_0^{\pi} x\,dx$$

$$= \frac{1}{\pi}\left[-\frac{x^2}{2}\right]_{-\pi}^{0} + \frac{1}{\pi}\left[\frac{x^2}{2}\right]_0^{\pi}$$

$$\Rightarrow \frac{1}{\pi}\left[-\frac{0}{2} + \frac{(-\pi)^2}{2}\right] + \frac{1}{\pi}\left[\frac{\pi^2}{2} - 0\right]$$

$$\Rightarrow \frac{\pi^2}{2\pi} + \frac{\pi^2}{2\pi}$$

$$\Rightarrow \frac{\pi}{2} + \frac{\pi}{2}$$

$$\Rightarrow \pi$$

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$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)\,dx$$

$$= \frac{1}{\pi}\int_{-\pi}^{0}(-x)\cos(nx)\,dx + \frac{1}{\pi}\int_{0}^{\pi} x\cos(nx)\,dx$$

$$= \frac{-1}{\pi}\int_{-\pi}^{0} x\cos(nx)\,dx + \frac{1}{\pi}\int_{0}^{\pi} x\cos(nx)\,dx$$

$$= -\frac{1}{\pi}\left[x\int\cos(nx)\,dx - \int 1\cdot\frac{\sin(nx)}{n}\,dx\right]_{-\pi}^{0} + \frac{1}{\pi}\left[x\int\cos nx\,dx - \int\left(1\cdot\frac{\sin(nx)}{n}\right)dx\right]$$

$$= -\frac{1}{\pi}\left[\frac{x\sin(nx)}{n} - \frac{1}{n}\int\sin(nx)\,dx\right]_{-\pi}^{0} + \frac{1}{\pi}\left[\frac{x\sin(nx)}{n} - \frac{1}{n}\int\sin(nx)\,dx\right]_{0}^{\pi}$$

$$= -\frac{1}{\pi}\left[x\frac{\sin(nx)}{n} - \frac{1}{n}\left\{\frac{-\cos(nx)}{n}\right\}\right]_{-\pi}^{0} + \frac{1}{\pi}\left[\frac{x\sin(nx)}{n} - \frac{1}{n}\cdot\left\{\frac{-\cos(nx)}{n}\right\}\right]_{0}^{\pi}$$

$$= -\frac{1}{\pi}\left[x\frac{\sin(nx)}{n} + \frac{\cos(nx)}{n^2}\right]_{-\pi}^{0} + \frac{1}{\pi}\left[\frac{x\sin(nx)}{n} + \frac{\cos(nx)}{n^2}\right]_{0}^{\pi}$$

$$= -\frac{1}{\pi}\left[\left(0+\frac{1}{n^2}\right) - \left(\frac{-\pi\sin(-n\pi)}{n} + \frac{\cos(-n\pi)}{n^2}\right)\right] + \frac{1}{\pi}\left[\left(\frac{\pi\sin(n\pi)}{n} + \frac{\cos(n\pi)}{n^2}\right) - \left(0+\frac{1}{n^2}\right)\right]$$

$$= -\frac{1}{\pi}\left[\frac{1}{n^2} - \left(0+\frac{(-1)^n}{n^2}\right)\right] + \frac{1}{\pi}\left[\left(0+\frac{(-1)^n}{n^2}\right) - \left(\frac{1}{n^2}\right)\right]$$

$$= -\frac{1}{n^2\pi} + \frac{(-1)^n}{n^2\pi} + \frac{(-1)^n}{n^2\pi} - \frac{1}{n^2\pi}$$

$$a_n = \frac{2(-1)^n}{n^2\pi} - \frac{2}{n^2\pi}$$

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$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(nx)\,dx$$

$$= \frac{1}{\pi}\int_{-\pi}^{0}(-x)\sin(nx)\,dx + \frac{1}{\pi}\int_{0}^{\pi} x\sin(nx)\,dx$$

| | |
|---|---|
| $-x$ | $\sin(nx)$ |
| $-1$ | $\frac{-\cos(nx)}{n}$ |
| $0$ | $\frac{-\sin(nx)}{n^2}$ |

$$= \frac{1}{\pi}\left[\frac{x\cos(nx)}{n} - \frac{\sin(nx)}{n^2}\right]_{-\pi}^{0} + \frac{1}{\pi}\left[-\frac{x\cos(nx)}{n} + \frac{\sin(nx)}{n^2}\right]_{0}^{\pi}$$

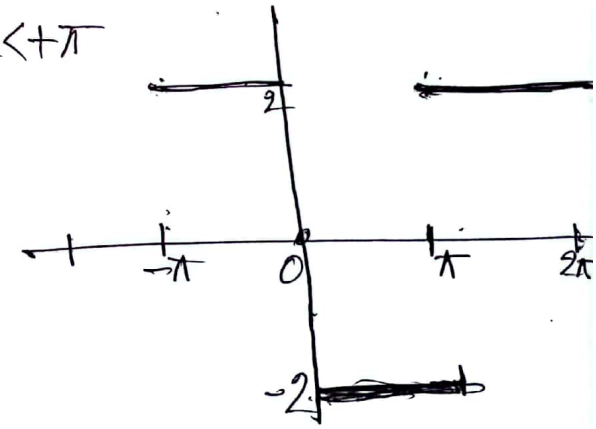
$$= \frac{1}{\pi}\left[(0-0) - \left(\frac{-\pi\cos(-n\pi)}{n} - \frac{\sin(-n\pi)}{n^2}\right)\right] + \frac{1}{\pi}\left[\left(\frac{-\pi\cos(n\pi)}{n} + \frac{\sin(n\pi)}{n^2}\right) - (0+0)\right]$$

$$= \frac{1}{\pi}\left[0 - \left(\frac{-\pi(-1)^n}{n} - 0\right)\right] + \frac{1}{\pi}\left[\frac{-\pi(-1)^n}{n} + 0 - 0\right]$$

$$= \frac{(-1)^n}{n} - (-1)^n = 0.$$

6. Draw sketches and determine the $a_0, a_n, b_n$ and therefore determine the Fourier series (upto five terms at least) for the following two functions:

$$f(x) = \begin{cases} -2 & \text{if } 0 < x < \pi \\ +2 & \text{if } \pi < x < 2\pi \end{cases} \quad \text{for } -\pi < x < +\pi$$




$f(x)$ is odd because:

$$f(x) = \begin{cases} -2 & 0 < x < \pi \\ +2 & -\pi < x < 0 \end{cases}$$

Let, $a = 2.6$

$f(-2.6) = f(-a) = 2$

$f(2.6) = f(a) = -2$

$f(-a) = -f(a) \rightarrow$ Odd.

$\therefore f(x) = \sum_{n=1}^{\infty} b_n \sin(nx) \quad [a_0 = 0,\ a_n = 0]$

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$$b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin(nx)\,dx$$

$$= \frac{2}{\pi}\int_0^{\pi}(-2)\sin(nx)\,dx$$

$$= \frac{-4}{\pi}\left[\frac{-\cos(nx)}{n}\right]_0^{\pi}$$

$$= \frac{4}{\pi}\left[\frac{\cos(nx)}{n}\right]_0^{\pi}$$

$$= \frac{4}{\pi}\left[\frac{\cos(n\pi)}{n} - \frac{1}{n}\right]$$

$$= \frac{4}{\pi}\left(\frac{(-1)^n}{n} - \frac{1}{n}\right)$$

$$= \frac{4\left[(-1)^n - 1\right]}{\pi n}$$

$$b_n = \frac{4\left[(-1)^n - 1\right]}{\pi n}$$

$b_1 = \frac{4}{\pi}(-1-1) = -\frac{8}{\pi}$

$b_2 = \frac{4}{2\pi}(1-1) = 0$

$b_3 = \frac{4}{3\pi}(-1-1) = -\frac{8}{3\pi}$

$b_4 = 0$

$b_5 = -\frac{8}{5\pi}$

$b_6 = 0$

$b_7 = \frac{-8}{7\pi}$

$b_8 = 0$

$b_9 = \frac{-8}{9\pi}$

So, the Fourier series (upto) five terms:

$$f(x) = \sum_{n=1}^{\infty} b_n \sin(nx)$$

$$= -\frac{8}{\pi}\sin(x) - \frac{8}{3\pi}\sin(3x) - \frac{8}{5\pi}\sin(5x) - \frac{8}{7\pi}\sin(7x) - \frac{8}{9\pi}\sin(9x) - \cdots$$

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Eulers Formula: $e^{i\theta} = \cos\theta + i\sin\theta$.

$e^{ix} = \cos x + i\sin x$.

$e^{inx} = \cos(nx) + i\sin(nx)$

## * Complex Notation for Fourier Series:

Using eulers identity:

$$e^{i\theta} = \cos\theta + i\sin\theta$$
$$e^{-i\theta} = \cos\theta - i\sin\theta$$

where $i$ is the imaginary unit such that $i^2 = -1$, the Fourier series for $f(x)$ can be written in complex form as:

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{(inx\pi/L)}$$

$$c_n = \frac{1}{2L}\int_{-L}^{L} f(x)\, e^{(-inx\pi/L)}\, dx$$

Proof:

$e^{i\theta} + e^{-i\theta} = 2\cos\theta$

$\Rightarrow \cos\theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$

$\therefore \cos\theta = \frac{1}{2}e^{i\theta} + \frac{1}{2}e^{-i\theta}$

$e^{i\theta} - e^{-i\theta} = 2i\sin\theta$

$\Rightarrow \sin\theta = \frac{1}{2i}e^{i\theta} - \frac{1}{2i}e^{-i\theta}$.

$\therefore \cos\left(\frac{nx\pi}{L}\right) = \frac{1}{2}e^{\left(\frac{inx\pi}{L}\right)} + \frac{1}{2}e^{\left(\frac{-inx\pi}{L}\right)}$

$\therefore \sin\left(\frac{nx\pi}{L}\right) = \frac{1}{2i}e^{\left(\frac{inx\pi}{L}\right)} - \frac{1}{2i}e^{\left(\frac{-inx\pi}{L}\right)}$

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We know,

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right)\right]$$

$$= \frac{a_0}{2} + \sum_{n=1}^{\infty}\left[a_n \frac{1}{2} e^{\left(\frac{in\pi x}{L}\right)} + a_n \frac{1}{2} e^{\left(\frac{-in\pi x}{L}\right)} + b_n \frac{1}{2i} e^{\left(\frac{in\pi x}{L}\right)} + -b_n \frac{1}{2i} e^{\left(\frac{-in\pi x}{L}\right)}\right]$$

$$= \frac{a_0}{2} + \sum_{n=1}^{\infty}\left[\left(\frac{1}{2}a_n + \frac{1}{2i}b_n\right) e^{\left(\frac{in\pi x}{L}\right)} + \left(\frac{1}{2}a_n - \frac{1}{2i}b_n\right) e^{\left(\frac{-in\pi x}{L}\right)}\right]$$

$$= \sum_{n=1}^{\infty}\left(\frac{1}{2}a_n - \frac{1}{2i}b_n\right) e^{\left(\frac{-in\pi x}{L}\right)} + \frac{a_0}{2} e^{\left(\frac{i.0\pi x}{L}\right)} + \sum_{n=1}^{\infty}\left(\frac{1}{2}a_n + \frac{1}{2i}b_n\right) e^{\left(\frac{in\pi x}{L}\right)}$$

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{\left(\frac{in\pi x}{L}\right)}$$

When,

$$c_n = \begin{cases} \frac{1}{2}a_n + \frac{1}{2i}b_n & \text{when } n>0 \\ \frac{a_0}{2} & \text{when } n=0 \\ \frac{1}{2}a_n - \frac{1}{2i}b_n & \text{when } n<0 \end{cases}$$

For a periodic function on $(-L, L)$,

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{\left(\frac{in\pi x}{L}\right)}$$

$$c_n = \frac{1}{2L}\int_{-L}^{L} f(x) e^{\left(\frac{-in\pi x}{L}\right)} dx \quad \text{——— (1)}$$

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Now, $f(x) = \sum_{n=-\infty}^{\infty} c_n e^{\left(\frac{in\pi x}{L}\right)}$

$\Rightarrow f(x) e^{\left(-\frac{im\pi x}{L}\right)} = \sum_{n=-\infty}^{\infty} c_n \cdot e^{\left(\frac{in\pi x}{L}\right)} \cdot e^{\left(-\frac{im\pi x}{L}\right)}$

$\Rightarrow \int_{-L}^{L} f(x) e^{\left(-\frac{im\pi x}{L}\right)} dx = \sum_{n=-\infty}^{\infty} c_n \int_{-L}^{L} e^{\left(\frac{in\pi x}{L}\right)} \cdot e^{\left(-\frac{im\pi x}{L}\right)} dx$ — (2)

By orthogonality:

$\int_{-L}^{L} e^{\left(\frac{in\pi x}{L}\right)} \cdot e^{\left(-\frac{im\pi x}{L}\right)} dx = 0$ if $m \neq n$

or $\int_{-L}^{L} e^{\left(\frac{in\pi x}{L}\right)} \cdot e^{\left(-\frac{im\pi x}{L}\right)} dx = 2L$ if $m = n$

$\Rightarrow \int_{-L}^{L} 1 \cdot dx = [x]_{-L}^{L} = 2L$ (m fixed)

From (2), $\int_{-L}^{L} f(x) e^{\left(-\frac{im\pi x}{L}\right)} dx = \sum_{n=-\infty}^{\infty} c_n \int_{-L}^{L} e^{\left(\frac{in\pi x}{L}\right)} \cdot e^{\left(-\frac{im\pi x}{L}\right)} dx$

$\Rightarrow \int_{-L}^{L} f(x) e^{\left(-\frac{im\pi x}{L}\right)} dx = c_n \cdot 2L$ [for $m = n$]

$\Rightarrow c_n = \frac{1}{2L} \int_{-L}^{L} f(x) e^{\left(-\frac{in\pi x}{L}\right)} dx$ (Proved)

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# Fourier Transform

Fourier Transform Equation (F.T.):

$$F(\omega) = \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx .$$

Inverse Fourier Transform (I.F.T.):

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega x} dx$$

9.1. (a) Find the Fourier transform of :

$$f(x) = \begin{cases} 1 & \text{for } |x| < a \\ 0 & \text{for } |x| > a . \end{cases}$$

(b) Use the result of (a) to evaluate $\int_{-\infty}^{\infty} \frac{\sin(\omega a)\cos(\omega x)}{\omega} d\omega$

(c) Deduce the value of $\int_0^{\infty} \frac{\sin(u)}{u} du$ .

**Soln.** (a) $F(\omega) = \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx = \int_{-1}^{1} 1 \cdot e^{-i\omega x} dx$ $\left| \begin{matrix} |x| < a \\ -a < x < a \\ 1 \end{matrix} \right.$

$$= \int_{-1}^{1} e^{-i\omega x} dx$$

$$= \left[ \frac{e^{-i\omega x}}{-i\omega} \right]_{-1}^{1}$$

$$= \left[ \frac{e^{-i\omega}}{-i\omega} - \frac{e^{i\omega}}{-i\omega} \right]$$

$$= \left[ \frac{e^{i\omega}}{i\omega} + \frac{e^{-i\omega}}{-i\omega} \right]$$

$$= \left[ \frac{e^{i\omega} - e^{-i\omega}}{i\omega} \right]$$

$$= \frac{2}{\omega} \cdot \frac{e^{i\omega} - e^{-i\omega}}{2i}$$

$$= \frac{2}{\omega} \cdot \sin\omega . \qquad \left[ \sin x = \frac{e^{ix} - e^{-ix}}{2i} \right]$$

Fourier Transform $\{f(x)\} = \frac{2}{\omega} \sin(\omega)$.

(b) $f(x) = I.F.T.\{F(\omega)\}$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega) e^{i\omega x} d\omega$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{2}{\omega}\sin\omega \cdot e^{i\omega x} d\omega.$$

$$= \frac{2}{2\pi}\int_{-\infty}^{\infty} \frac{\sin(\omega)}{\omega}\left[\cos(\omega x) + i\sin(\omega x)\right] d\omega \quad \left[e^{i\theta} = \cos\theta + i\sin\theta\right]$$

$$f(x) = \underbrace{\frac{1}{\pi}\int_{-\infty}^{\infty} \frac{\sin(\omega)}{\omega}\cdot\cos(\omega x) d\omega}_{\text{Real Part}} + \underbrace{i\,\frac{1}{\pi}\int_{-\infty}^{\infty} \frac{\sin\omega}{\omega}\sin(\omega x) d\omega}_{\text{Imaginary part} = 0}$$

[For this Math]

$$f(x) = \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{\sin\omega}{\omega}\cos(\omega x) d\omega \quad \text{(Ans)}$$

(c) $\frac{1}{\pi}\int_{-\infty}^{\infty} \frac{\sin\omega}{\omega}\cos(\omega x) d\omega = f(x)$

Set $\omega = u$ and $x = 0$ :

$$\frac{1}{\pi}\int_{-\infty}^{\infty} \frac{\sin u}{u}\cos(0)\, du = f(0)$$

$$\Rightarrow \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{\sin u}{u} du = f(0)$$

$$\Rightarrow \int_{-\infty}^{\infty} \frac{\sin(u)}{u} du = f(0)\cdot\pi$$

$$\Rightarrow 2\int_{0}^{\infty} \frac{\sin(u)}{u} du = 1\cdot\pi \quad [f(0) = 1 \text{ Here}]$$

$$\therefore \int_{0}^{\infty} \frac{\sin(u)}{u} du = \frac{\pi}{2} \quad \text{(Ans)}$$

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② (a) Find the Fourier Transform of:

$$f(x)=\begin{cases}1-x^2 & \text{when } |x|<1\\ 0 & \text{when } |x|>1.\end{cases}$$

(b) Use the result in part (a) to evaluate:

$$\int_0^\infty \left(\frac{x\cos x-\sin x}{x^3}\right)\cos\frac{x}{2}\,dx.$$

**Soln** (a) $F(w)=\int_{-\infty}^{\infty} f(x)e^{-iwx}\,dx$

$$=\int_{-1}^{1}(1-x^2)e^{-iwx}\,dx$$

$$=\left[(1-x^2)\frac{e^{-iwx}}{(-iw)}-(-2x)\frac{e^{-iwx}}{(-w^2)}+(-2)\frac{e^{-iwx}}{(-i)w^3}\right]_{-1}^{1}$$

$$=\left[\left(0-2\cdot\frac{e^{-iw}}{w^2}-2\cdot\frac{e^{-iw}}{iw^3}\right)-\left(0+2\cdot\frac{e^{iw}}{w^2}-2\cdot\frac{e^{iw}}{iw^3}\right)\right]$$

$$=-\frac{2}{w^2}e^{-iw}-\frac{2}{w^3 i}e^{-iw}-\frac{2}{w^2}e^{iw}+\frac{2}{w^3 i}e^{iw}$$

$$=-\frac{2}{w^2}\left(e^{-iw}+e^{iw}\right)+\frac{2}{w^3}\left(\frac{e^{iw}-e^{-iw}}{i}\right)$$

$$=-\frac{2\cdot2}{w^2}\left(\frac{e^{-iw}+e^{iw}}{2}\right)+\frac{2\cdot2}{w^3}\left(\frac{e^{iw}-e^{-iw}}{2i}\right)$$

$$=-\frac{4}{w^2}\cos w+\frac{4}{w^3}\sin w.$$

$$=\frac{4\sin w-4w\cos w}{w^3}$$

$|x|<1$
$\Rightarrow -1<x<1$

$$\int_{-\infty}^{-1}0\,dx+\int_{-1}^{1}(1-x^2)e^{-iwx}\,dx+\int_{1}^{\infty}0\,dx$$

| | |
|---|---|
| $1-x^2$ | $e^{-iwx}$ |
| $-2x$ | $\frac{e^{-iwx}}{-iw}$ |
| $-2$ | $\frac{e^{-iwx}}{i^2w^2}$ |
| $0$ | $\frac{e^{-iwx}}{i^3w^3}$ |

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(b) $f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{4\sin w - 4w\cos w}{w^3} e^{iwx}\, dw.$

$= \frac{2}{2\pi} \int_{-\infty}^{\infty} \left\{\frac{\sin w - w\cos w}{w^3}\right\} \{\cos(wx) + i\sin(wx)\}\, dw$

$f(x) = \underbrace{\frac{2}{\pi} \int_{-\infty}^{\infty} \frac{\sin w - w\cos w}{w^3} \cdot \cos(wx)\, dx}_{\text{Real}} + i \cdot \underbrace{\frac{2}{\pi} \int_{-\infty}^{\infty} \frac{\sin w - w\cos w}{w^3} \cdot \sin(wx)\, dw}_{0 = \text{Imaginary}}$

$f(x) = \frac{2}{\pi} \int_{-\infty}^{\infty} \left\{\frac{\sin w - w\cos w}{w^3}\right\} \cos(wx)\, dw.$

$\Rightarrow f\left(\frac{1}{2}\right) = \frac{2}{\pi} \int_{-\infty}^{\infty} \left\{\frac{\sin w - w\cos w}{w^3}\right\} \cos\left(\frac{1}{2}w\right) dw.$

$\Rightarrow f\left(\frac{1}{2}\right) = \frac{2}{\pi} \int_{-\infty}^{\infty} \frac{\sin x - x\cos x}{x^3} \cdot \cos\left(\frac{1}{2}x\right) dx = \frac{3}{4}$

$\Rightarrow \int_{-\infty}^{\infty} \frac{\sin x - x\cos x}{x^3} \cdot \cos\left(\frac{1}{2}x\right) dx = \frac{3}{4} \cdot \frac{\pi}{2}$ $\left[f\left(\frac{1}{2}\right) = 1 - \left(\frac{1}{2}\right)^2 = 1 - \frac{1}{4} = \frac{3}{4}\right]$

$\Rightarrow \int_{-\infty}^{\infty} \frac{x\cos x - \sin x}{x^3} \cdot \cos\left(\frac{1}{2}x\right) dx = -\frac{3\pi}{8}$

$\Rightarrow \int_{0}^{\infty} \frac{x\cos x - \sin x}{x^3} \cdot \cos\left(\frac{1}{2}x\right) dx = \left(-\frac{3\pi}{8}\right) \times \frac{1}{2}$

$\therefore \int_{0}^{\infty} \frac{x\cos x - \sin x}{x^3} \cdot \cos\left(\frac{1}{2}x\right) dx = -\frac{3\pi}{16}.$

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## *Fourier Sine and Cosine Transform:

### *Fourier Sine Transform:

$$F_S(\omega) = \int_0^\infty f(x) \sin(\omega x)\, dx$$

$$f(x) = \frac{2}{\pi} \int_0^\infty F_S(\omega) \sin(\omega x)\, d\omega$$

### *Fourier Cosine Transform:

$$F_C(\omega) = \int_0^\infty f(x) \cos(\omega x)\, dx$$

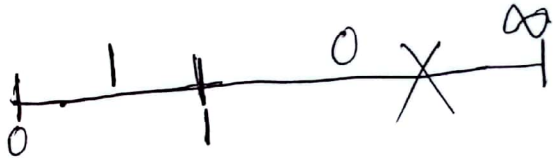
$$f(x) = \frac{2}{\pi} \int_0^\infty F_C(\omega) \cos(\omega x)\, d\omega.$$

Q.1. If $\boxed{f(x) = \begin{cases} 1 & \text{for } 0 \le x < 1 \\ 0 & \text{for } x \ge 1 \end{cases}}$, Find:

(a) The Fourier Sine Transform of $f(x)$.

(b) The Fourier Cosine Transform of $f(x)$.

Sol$^n$. (a) $F_S(\omega) = \int_0^\infty f(x) \sin(\omega x)\, dx$




$$F_S(\omega) = \int_0^1 1 \cdot \sin(\omega x)\, dx + \int_1^\infty 0 \cdot \sin(\omega x)\, dx$$

$$= \int_0^1 \sin(\omega x)\, dx$$

$$= \left[\frac{-\cos(\omega x)}{\omega}\right]_0^1$$

$$= \frac{-\cos(\omega)}{\omega} - \frac{-\cos(0)}{\omega}$$

$$= \frac{1}{\omega} - \frac{\cos(\omega)}{\omega}$$

$$= \frac{1 - \cos(\omega)}{\omega}$$ (Ans)

(b) $F_c(\omega) = \int_0^\infty f(x)\cos(\omega x)\,dx$

$= \int_0^1 1.\cos(\omega x)\,dx + \int_1^\infty 0.\cos(\omega x)\,dx$

$= \int_0^1 \cos(\omega x)\,dx$

$= \left[\frac{\sin(\omega x)}{\omega}\right]_0^1$

$= \left[\frac{\sin(\omega)}{\omega} - \frac{\sin(0)}{\omega}\right]$

$\therefore F_c(\omega) = \frac{\sin(\omega)}{\omega} - 0 = \frac{\sin(\omega)}{\omega}$ (Ans)

Q. (a) Find the Fourier sine transform of $e^{-x}$ $[x \geq 0]$

(b) Show that: $\int_0^\infty \frac{x\sin(mx)}{x^2+1}\,dx = \frac{\pi}{2}e^{-m}$ $[m>0]$,

by using the result in (a)

Sol$^{n}$ (a) $F_s(\omega) = \int_0^\infty f(x)\sin(\omega x)\,dx$

$I = \int_0^\infty e^{-x}\sin(\omega x)\,dx$

$= \left[\frac{e^{-x}((-1)\sin\omega x - \omega\cos\omega x)}{(-1)^2+\omega^2}\right]_0^\infty$

$= \left[\frac{e^{-x}(-\sin\omega x - \omega\cos\omega x)}{1+\omega^2}\right]_0^\infty$

$= 0 - \left(\frac{e^0(0-\omega\cdot 1)}{1+\omega^2}\right) = \frac{\omega}{1+\omega^2}$ (Ans)

(1) $\int e^{ax}\cos(bx)\,dx = \frac{e^{ax}(a\cos bx + b\sin bx)}{a^2+b^2}$

(2) $\int e^{ax}\sin(bx)\,dx = \frac{e^{ax}(a\sin bx - b\cos bx)}{a^2+b^2}$

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(b) $f(x) = \frac{2}{\pi} \int_0^\infty \frac{w}{1+w^2} \cdot \sin(wx)\, dw.$

$\Rightarrow f(m) = \frac{2}{\pi} \int_0^\infty \frac{w}{1+w^2} \sin(mw)\, dw.$

$\Rightarrow f(m) = \frac{2}{\pi} \int_0^\infty \frac{x}{1+x^2} \cdot \sin(mx)\, dx$

$\Rightarrow \int_0^\infty \frac{x \sin(mx)}{x^2+1} dx = \frac{\pi}{2} f(m) \qquad \left[f(x) = e^{-mx}\right]$

$= \frac{\pi}{2} e^{-m}$ (Ans)

Q.2(a) Find the Fourier cosine transform of

$f(x) = e^{-mx} \quad [m>0]$

(b) Use the result in (a) to show that:

$\int_0^\infty \frac{\cos pv}{v^2+p^2} dv = \frac{\pi}{2p} e^{-pp} \quad (p>0, p>0)$

Soln: $F_c(w) = \int_0^\infty e^{-mx} \cos(wx)\, dx$

$I = \int_0^\infty e^{-mx} \cos(wx)\, dx$

$= \left[e^{-mx} \frac{\sin(wx)}{w}\right]_0^\infty - \int_0^\infty (-m) e^{-mx} \cdot \frac{\sin wx}{w} dx$

$= 0 - 0 + \frac{m}{w} \int_0^\infty e^{-mx} \sin(wx)\, dx$

$= \frac{m}{w} \left( \left[e^{-mx} \cdot \frac{-\cos(wx)}{w}\right]_0^\infty - \int_0^\infty (-m) e^{-mx} \cdot \frac{-\cos wx}{w} dx \right.$

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$$= \frac{m}{w}\left(0 - \frac{-1}{w}\right) - \frac{m^2}{w^2}\int_0^\infty e^{-mx}\cdot\cos wx\,dx$$

$$\therefore I = -\frac{m}{w^2} - \frac{m^2}{w^2} I$$

$$\Rightarrow w^2 I = +m - m^2 I$$

$$\Rightarrow I = \frac{+m}{m^2+w^2}.$$

$$\therefore f(x) = \frac{2}{\pi}\int_0^\infty \frac{m}{m^2+w^2}\cos(wx)\,dw. \quad (\text{A})$$

(b) Show: $\boxed{\int_0^\infty \frac{\cos(pv)}{v^2+\beta^2}\,dv = \frac{\pi}{2\beta}e^{-p\beta}}$.

$$f(x) = \frac{2}{\pi}\int_0^\infty \frac{m}{m^2+w^2}\cos(wx)\,dw.$$

Set $x=p$, $w=v$, $m=\beta$ and:

$$f(p)\cdot\frac{\pi}{2} = \int_0^\infty \frac{\beta\cos(pv)}{\beta^2+v^2}\,dv.$$

$$\Rightarrow e^{-\beta p}\cdot\frac{\pi}{2} = \beta\int_0^\infty \frac{\cos(pv)}{\beta^2+v^2}\,dv$$

$$\therefore \int_0^\infty \frac{\cos(pv)}{v^2+\beta^2}\,dv = \frac{\pi}{2\beta}e^{-\beta p}. \quad (\text{M})$$

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